

0.1 Why the Response Remains of the Closed Form: Proposition, Bound, and Verification

0.1.1 Exact dynamics and the structure of the perturbations

About the active pivot line P , the exact roll dynamics during ground contact read (the pitch case is analogous)

$$J_P \dot{\omega} = M_{x,P}(t) + \Delta M_{GE,P}(t, \phi) - W(l_p + \lambda_{\text{off}}) \cos \phi + W z_{\text{CoM}} \sin \phi, \quad (1)$$

with $M_{x,P} = M_x + f l_p$ and, conservatively, $l_{p,\phi} = 0.140$ m, $l_{p,\theta} = 0.110$ m. The ground-contact phase is a one-degree-of-freedom rigid rotation about the gear line, so every rotor height is an exact trigonometric function of the single angle ϕ ; in the level configuration all heights coincide, yielding the exact identity

$$\Delta M_{GE,P} = \eta(\phi) (M_x + f l_p), \quad \eta(\phi) \triangleq \bar{\gamma}(\phi) - 1, \quad (2)$$

where $\bar{\gamma}$ denotes the rotor-common thrust factor. The identity is *independent of the CoM position* — λ_{off} enters (1) only through the gravity arm — so ground effect biases the identified critical moment identically for every CoM configuration. The magnitude is evaluated with two models of increasing physics content. The per-rotor superposition of the single-rotor Cheeseman–Bennett result (which *neglects* rotor–rotor interference) gives, at the measured geometry ($h = 0.315$ m, $R = 0.127$ m), $\eta(0) = 1.03\%$ independent of l_p . The reported model adds the interference from first principles by the image method: each rotor is a three-dimensional potential source of strength $R^2 v_\infty / 4$, the ground plane is enforced by mirror images, and only the vertical projection of each image-induced velocity reduces the downwash, giving the per-rotor sum

$$s_{\text{rot},i} = \frac{R^2}{4} \sum_j \frac{z_i + z_j}{[d_{ij}^2 + (z_i + z_j)^2]^{3/2}}, \quad \gamma_i = (1 - s_{\text{rot},i})^{-1}, \quad (3)$$

evaluated at each rotor's *own* tilted height $z_i = h \cos \phi + s_i \sin \phi$ — no averaging enters, and the attitude dependence of the model resides entirely in this term. The self term $j = i$ reduces exactly to Cheeseman–Bennett, so (3) contains the single-rotor result as the zero-neighbour limit and adds *no empirical constant of any kind*: the neighbour sum is fixed by the measured rotor geometry alone. Mapping the per-rotor gains to the pivot,

$$\Delta M_{GE} = \sum_i b_i (\gamma_i - 1) T_i + \left[\sum_i (\gamma_i - 1) T_i \right] l_p, \quad (4)$$

with b_i the body-centre arms and the second term the parallel-axis transfer of the net vertical surplus. Under constant total thrust the result is affine in the commanded moment, $\Delta M_{GE} = a(\phi) + b(\phi) M_{\text{cmd}}$, with fitted coefficients $a/(f l_p) [\%] = 4.27 - 0.086 \phi$ and $b [\%] = 4.28 - 0.060 \phi$ (ϕ in degrees) — a factor ≈ 4 above the interference-free value in both channels, which is the quantitative statement that rotor–rotor interference is not negligible at this clearance. The affine decomposition, the anti-restoring sign of the height-asymmetry moment, and the channel structure below are identical in the two models — they follow from thrust-proportionality and smooth height dependence alone; only the magnitudes differ.

How far does this move the calibrated constants? With $k_0^{\text{GE}} \triangleq \beta_f f l_p + \beta_M M_{\text{crit}}$ the onset value of the ϕ -slope, the effective anti-restoring gradient becomes $D = W z_{\text{CoM}} + k_0^{\text{GE}}$ ($k_0^{\text{GE}} < 0$: the ground effect *opposes* the tilt and therefore reduces D). Since $C_2 = \sqrt{D/J_P}$, the inertia cancels in the ratio and no inertia estimate is needed:

$$\frac{\Delta C_2}{C_2} = \sqrt{1 - \frac{|k_0^{\text{GE}}|}{W z_{\text{CoM}}}} - 1, \quad \frac{\Delta K}{K} = \frac{1 + \alpha_M}{1 - |k_0^{\text{GE}}|/(W z_{\text{CoM}})} - 1. \quad (5)$$

Over the operating box ($W z_{\text{CoM}} \in [6.3, 9.5]$ N m for $z_{\text{CoM}} \in [0.2, 0.3]$ m, measured onset thrust $f \approx 21$ N, $|M_{\text{crit}}| \leq 2.1$ N m) the interference model gives $|k_0^{\text{GE}}| \leq 0.217$ N m/rad (roll) and 0.172 (pitch), i.e. at most 3.4% and 2.7% of $W z_{\text{CoM}}$, so

$$\frac{\Delta C_2}{C_2} \geq -1.7\% \text{ (roll)}, \quad -1.4\% \text{ (pitch)}, \quad \frac{\Delta K}{K} \leq +8.0\%,$$

the amplitude constant moving more because it carries the moment-channel gain α_M and the gradient reduction together. Two consequences. First, a *single* C_2 per configuration remains legitimate: within one dataset the only run-to-run variation of k_0^{GE} enters through M_{crit} , whose largest directional spread (0.96 N m) changes D by 0.52% and C_2 by 0.26%. Second, these shifts are *absorbed*, not committed as errors — the calibration estimates the effective constants directly — and the sensitivity analysis shows

that a $\pm 20\%$ mis-specification of either constant moves the identified CoM offset by less than 0.9 mm, an order of magnitude of slack above the numbers above. For scale, the a priori uncertainty of C_2 from the z_{CoM} box alone is -12.5% ($6.14 \rightarrow 5.37$ rad/s with $J_P \geq m(z_{\text{CoM}}^2 + l_p^2)$), several times the ground-effect shift — which is precisely why (C_2, K) are calibrated rather than computed.

Everything that follows is deliberately insensitive to that magnitude. The absorption argument assumes only smoothness and thrust-proportionality, which any near-ground aerodynamic model shares, and the same is true of a purely geometric alternative: a contact-force lever displaced inboard of the kinematic pivot circle by a finite, compliant contact patch enters (1) in the same functional position as the a -channel. The static check of the identified thresholds indeed cannot separate the two — a 19.2 ± 1.8 mm inboard lever reproduces the measured deficit as well as the interference model does (residual RMS 82.3 versus 86.0 mN m), the discriminating moment-proportional channel being below the group-to-group scatter. This is a limitation of the attribution, not of the method: the bounds below hold for either cause, and both cancel antisymmetrically in the pivot-free deliverable.

0.1.2 Ideal configuration: the cosh family and its escape channel

Consider first the ideal level configuration: the vehicle rests at $\phi = 0$, so the onset — defined dynamically by zero net moment, $\omega(0) = \dot{\omega}(0) = 0$ — occurs at $\phi = 0$, and the operating point $(\tau, \phi, T) = (0, 0, T^*)$ makes the onset expansion and the textbook small-angle expansion one and the same. The tilt variable and the excursion coincide, $\delta\phi = \phi$.

Linearizing (1) about this point collects every perturbation into three absorbable channels plus a remainder: constants are absorbed by the onset anchoring; terms $\propto \dot{M}\tau$ (including the thrust-ramp channel of (2), whose coefficient η^* is rate-independent) are absorbed by the ramp-invariance calibration of K ; terms $\propto \delta\phi$ (the gravity slope Wz_{CoM} — here an *anti-restoring* gradient, i.e. a negative stiffness, which is what makes the response hyperbolic rather than oscillatory — and $\eta' M_P^*$) are absorbed by the calibrated C_2 . What escapes is of two kinds — the second-order Taylor remainder of the gravity terms and, from the ground effect, the one product that is bilinear in two spanned coordinates:

$$\rho = \underbrace{\frac{1}{2} G''(\xi) \delta\phi^2}_{\text{gravity curvature}} + \underbrace{\beta_M \dot{M} \tau \delta\phi}_{\text{ground effect}} + \dots, \quad G''(\xi) = W(l_p + \lambda_{\text{off}}) \cos \xi - Wz_{\text{CoM}} \sin \xi, \quad (6)$$

the second term arising because the moment-proportional ground-effect coefficient is itself attitude dependent: writing $b(\phi) = \alpha_M + \beta_M \delta\phi$ and $M_{\text{cmd}} = M_{\text{crit}}^* + \dot{M}\tau$, the product $b(\phi)M_{\text{cmd}}$ splits into a constant, a term $\propto \dot{M}\tau$, a term $\propto \delta\phi$ with coefficient $\beta_M M_{\text{crit}}^*$ — all three absorbed above — and this single bilinear leftover. It is quantified in Sec. 0.1.3 and enters Table 1 on its own row. With ξ between 0 and ϕ in the first term: the sine term enters at second order *with a negative sign* and is dropped conservatively, $|G''| \leq W(l_p + \lambda_{\text{off}})$. At the 5° excursion this bound evaluates to 19.2 mN m (roll) / 15.6 mN m (pitch).

Proposition 1 *Under (i) maintained pivot contact, (ii) gate-passing ramp execution, and (iii) any thrust-proportional, smooth ground-effect model, subtracting the nominal (onset-balanced) equation from (1) yields for the deviation $e_\phi \triangleq \phi - \phi_{\text{nom}}$*

$$J_P \ddot{e}_\phi - Wz_{\text{CoM}} e_\phi = \rho(\tau), \quad e_\phi(0) = \dot{e}_\phi(0) = 0, \quad (7)$$

where the state-dependent remainder is recast as an equivalent exogenous input, $\rho(\tau) \triangleq \rho(\delta\phi(\tau))$, and $Wz_{\text{CoM}} = J_P C_2^2$ is understood as the effective anti-restoring gradient that the calibrated C_2 estimates. Every solution of the unforced problem with the onset conditions $\omega(0) = \dot{\omega}(0) = 0$ is exactly $\omega(\tau) = C_1(\cosh C_2\tau - 1)$ with $C_1 = KM$ fixed — no shape parameter is free. Ground effect and the gravity nonlinearity therefore move the system within the two-parameter cosh family; the only escape channel is ρ .

Impulse response and the propagated bound. The response of (7) to a unit moment impulse $\rho = \delta(\tau)$ follows from the momentum jump $J_P[\dot{e}_\phi(0^+) - \dot{e}_\phi(0^-)] = 1$ (the position cannot jump) and the unstable homogeneous solution: $e_\phi = \sinh(C_2\tau)/(J_P C_2)$, hence the rate impulse response $h_\omega(\tau) = \dot{e}_\phi = \cosh(C_2\tau)/J_P$ — consistent with $h_\omega(0) = 1/J_P$ from the jump condition. Superposition (Duhamel's integral [ref]) then gives the exact solution for arbitrary ρ and, normalizing like the empirical fit metric by the window-edge signal, the finite monotone envelope

$$\delta\omega(\tau) = \frac{1}{J_P} \int_0^\tau \cosh(C_2(\tau-s)) \rho(s) ds, \quad \frac{|\delta\omega(\tau)|}{\omega_{\text{nom}}(\tau_{\text{end}})} \leq \frac{C_2 \rho_{\text{max}}}{\dot{M}} \frac{\sinh(C_2\tau)}{\cosh(C_2\tau_{\text{end}}) - 1}, \quad (8)$$

with $\rho_{\max} \triangleq \sup_{[0, \tau_{\text{end}}]} |\rho|$, attaining $(C_2 \rho_{\max} / \dot{M}) \coth(C_2 \tau_{\text{end}} / 2)$ at the edge. Three points sharpen the bound. (a) *Kernel-forcing separation*. The onset conditions give the coherent forcing a sixth-order zero at the anchor ($\omega_{\text{nom}} \sim \tau^2$, hence $\delta\phi \sim \tau^3$ and $\rho \sim \delta\phi^2 \sim \tau^6$, leading order for $C_2 \tau \lesssim 1$), which is exactly where the unstable kernel amplifies most; conversely the forcing reaches $\rho_{\max}$ only at the window edge, where the kernel is unity. At the window midpoint the forcing is below 1% of its edge value. The quoted deviations are therefore *not* obtained from the uniform bound: the exact nominal remainder profile,

$$\rho(s) = \frac{1}{2} G_2 \left[\frac{C_1}{C_2} (\sinh C_2 s - C_2 s) \right]^2, \quad G_2 = W(l_p + \lambda_{\text{off}}) (1 + 5\%_{\text{GE}}), \quad (9)$$

is convolved with the kernel by direct numerical quadrature of (8) — no series truncation; the τ^6 law is only the small-argument limit, and the exact profile is in fact *more* edge-concentrated than the power law, its tail growing exponentially. This exact propagation tightens the uniform bound by a factor 25–96; a hand-derivable version splits the window at its midpoint, where the early half carries less than 1/128 of the forcing mass while the late-half kernel stays below $\cosh(C_2 \tau_{\text{end}} / 2)$, squeezing the convolution between 1 and ≈ 2.2 times the plain integral of ρ . Since (9) enters *without* first removing its span-absorbable part, the result is doubly conservative. (b) *State dependence*. A priori the remainder is evaluated along the nominal trajectory; the small-gain closure with $L = \max |d\rho/d\phi| \approx 0.95 \text{ N m/rad} \ll W z_{\text{CoM}}$ costs a factor $(1 - L/W z_{\text{CoM}})^{-1} \leq 1.11$. (Along the measured trajectory, Sec. 0.1.3, the bound is unconditional.) (c) *Window length*. Since $\delta\phi(\tau) = (C_1/C_2)(\sinh C_2 \tau - C_2 \tau)$ is strictly monotone ($d\delta\phi/d\tau = \omega_{\text{nom}} > 0$), the angle threshold determines τ_{end} bijectively: $C_2 \tau_{\text{end}} \in [2.1, 3.9]$ over the candidate rates, hence $\coth(C_2 \tau_{\text{end}} / 2) \in [1.04, 1.28]$ and the bound is effectively time-free, $|\delta\omega/\omega| \leq 1.3 C_2 \rho_{\max} / \dot{M}$.

Design consequence: the excitation cutoff and rate set. Because the propagated deviation is a computable function of the ramp rate and the excursion cap alone, the ideal analysis *selects* the excitation design. The systematic deviation grows with the cap and is worst at the slowest rate, crossing 1% at 4.8° for 0.10 N m/s and only at 8.9° for 1.20 N m/s; conversely the noise-limited residual floor falls with the cap (13% at 3°, 8% at 5°) and the post-onset sample count exceeds 40 everywhere. The 5° attitude cutoff is therefore the largest common cap that keeps the coherent deviation at the 1% level for every candidate rate, the binding constraint being the slowest ramp. In absolute terms the propagated deviation is 3.1–5.2 mrad/s at the 5° cap and at most 23 mrad/s at 9–10° — an order of magnitude below the in-motion gyro noise ($\approx 35 \text{ mrad/s RMS}$).

0.1.3 Transfer to the experimental configuration

The experiment departs from the ideal configuration in one respect: the sloped test surface places the rest attitude — and hence the onset, since the vehicle does not rotate before it — at $\phi^* \leq 2.6^\circ$ (median 1.7°, measured over all 140 runs). The correct operating point is $(0, \phi^*, T^*)$, the expansion variable becomes the excursion $\delta\phi = \phi - \phi^*$ (measured: at most 9.4° over the fit window, median 6.1°), and every operating-point effect transfers with an explicit bound:

- the effective anti-restoring gradient shifts from $W z_{\text{CoM}}$ to $W z_{\text{CoM}} \cos \phi^* + W(l_p + \lambda_{\text{off}}) \sin \phi^*$ — at most +2.3%, absorbed by the calibrated C_2 ;
- the onset-instant small-angle bias grows from zero to $\frac{1}{2} W(l_p + \lambda_{\text{off}}) \phi^{*2} \leq 5.4 \text{ mN m}$ per direction — common to both tip directions, which start from the same rest attitude, hence cancelling in M_{ff} ;
- the Lagrange remainder is in fact *smaller* at the tilted point (18.0 vs. 19.2 mN m at the 5° excursion), the restoring term subtracting from the curvature; the design tables, functions of the excursion alone, apply unchanged.

The slope’s genuine effect is geometric, not modelling: it renders the two pitch directions asymmetric in runway ($\approx 3^\circ$ vs. 7° to the absolute cutoff), lowering the SNR of the negative-pitch runs — precisely the regime for which the physics-constrained detector is needed.

On the experimental side every component of ρ is also *measured*: the recorded attitude and rotor speeds are evaluated through the exact expressions and projected out of the estimator span $\{1, \tau, \delta\phi\}$ — an algebraic identity, these being the estimator’s exact per-run degrees of freedom. Table 1 compares the ideal-theory bounds with the measured values; the measured residuals, on which the claims rest, lie consistently below their a priori counterparts. For the ground effect the projection is evaluated with and without rotor–rotor interference: the measured slope ratio is +0.99% of $\dot{M}$ (interference neglected) and +4.20% (interference included, IQR [4.13, 4.27]%), *both* rate-invariant across the twelvefold rate

Table 1: Components of the residual forcing ρ over the fit window. “A priori” bounds are the ideal-configuration values of Sec. 0.1.2 evaluated at the measured tilt range; “measured” values are evaluated along the recorded trajectories and projected out of the estimator span $\{1, \tau, \delta\phi\}$. All values in mN m.

Component	A priori	Measured (RMS / max)	Character
Small-angle curvature	19.2 at 5° ; ≈ 28 at the median excursion	1.2 / 12.3	coherent, edge-concentrated
Ground effect, without / with rotor interference	3.2 / 7.5 (tangent deviation, full range)	0.16/2.3 $\rightarrow$ 0.67/9.3	coherent, edge-concentrated
Bilinear $\tau \delta\phi$ (GE moment channel)	3.9 (envelope of the factor maxima)	0.04 / 0.17	coherent, almost fully absorbed
Ramp nonlinearity	30 (gate, full window)	1.6 / 6.2 (fit segment)	incoherent (noise-like)

range — the property the K -calibration absorption requires — and the out-of-span residual grows from 0.16 / 2.3 to 0.67 / 9.3 mN m (median RMS / max) with interference, while remaining edge-concentrated, so the propagated shape deviation stays at the few-percent level, below the noise-dominated residual. One term of the affine decomposition deserves explicit accounting. Because the moment coefficient is itself attitude dependent, the product $b(\phi)M_{\text{cmd}}$ contains $\beta_M \dot{M} \tau \delta\phi$ — a bilinear term in two spanned coordinates, hence the one piece of the ground-effect moment that is *not* in $\{1, \tau, \delta\phi\}$. Evaluated along the measured trajectories with $\beta_M = 0.0345 \text{ rad}^{-1}$ (`analysis/ge_bilinear.py`), its raw peak is 3.0 mN m and what survives the projection is 0.17 mN m at most (median 0.04) — the product is strongly correlated with the spanned coordinates along real trajectories, so it is absorbed almost entirely. It needs no separate treatment: propagated through the same Duhamel integral as every other component — and conservatively, using the *unprojected* forcing — it deforms the rate by 0.04% (slowest ramp) to 0.09% (fastest) of the nominal, against 0.8–1.9% for the small-angle curvature over the same windows. Its smallness has a structural reading: since $\varphi \approx \frac{1}{6}C_1 C_2^2 \tau^3$ near the onset, the term grows as τ^4 and is therefore edge-concentrated where the kernel has no time left to amplify it; equivalently, $\beta_M \dot{M} \tau$ is a *time-varying increment of the φ -coefficient*, i.e. the effective anti-restoring gradient drifts by $\beta_M \dot{M} \tau_{\text{end}} / D \leq 0.19\%$ across the window — a drift that small produces a response all but indistinguishable from the fixed-gradient cosh, which is why the projection removes 94% of it. As a state-dependent term it also enters the small-gain constant, $\partial\rho/\partial\varphi = \beta_M \dot{M} \tau \leq 0.021 \text{ N m/rad}$, raising L by 2.2% and leaving the closure factor unchanged to two digits. The channel fates are model-independent: the thrust channel (rotor-common and body lift alike) acts through the arm l_p exactly, is antisymmetric between tip directions, and cancels in M_{ff} regardless of its magnitude; the moment coefficient leaves a relative bias on M_{ff} between -1.0% (no interference) and -4.3% (with interference), i.e. 3–14 mN m or 0.10–0.44 mm of CoM offset — below the validation scatter (52 mN m RMS), which is dominated by the genuine contact asymmetries the pivot-free average is designed to capture, and present identically at lift-off, where M_{ff} compensates it by design. The full chain of reductions is $85 \rightarrow 28 \rightarrow 12.3 \rightarrow 1.2$ mN m: the unconditional $\phi=0$ envelope at the largest absolute tilt (10°), the onset-anchored Lagrange bound at the median measured excursion, the measured pointwise maximum after projection, and the measured window RMS.

0.1.4 Empirical verification: a prediction test, not a fit

Because the constrained estimator has no free shape parameter per run — $C_1 = K\dot{M}$ with $\dot{M}$ measured, C_2 shared, baseline fixed by continuity — the post-onset curve is fully determined before the post-onset data are seen; goodness of fit is evidence, not flexibility. With $\text{NRMSE} = [\frac{1}{N} \sum_k (\omega_k - \hat{\omega}_k)^2]^{1/2} / \max_k |\omega_k - C|$, four measured, model-independent certificates close the claim:

1. *Rate invariance.* Across all 140 gate-passing runs the NRMSE is 6.3% (median; $R^2 = 0.95$) and flat from 0.10 to 1.20 N m/s (5.5–6.6%): a state-dependent deformation would grow with amplitude, and none does. The residual sits at $2.2\times$ the quiet pre-onset gyro floor — motion- and thrust-induced measurement noise — while the predicted systematic deviation ($\leq 1\%$) lies below it.
2. *Trajectory-evaluated forcing.* The small-angle forcing deviates from the absorbable span by at most 12.3 mN m (median RMS 1.2), edge-concentrated where the kernel provides no amplification; the ground-effect forcing by 2.3 to 9.3 mN m depending on whether rotor interference is included. The

measured slope ratios, +0.99% of $\dot{M}$ (without interference) and +4.20% (with), reproduce each model's coefficient in situ and are rate-independent in both cases — the property on which the K -calibration absorption rests.

3. *Window ablation.* Truncating the window at a 3° relative excursion — where the small-angle model is unimpeachable — changes the per-direction critical moments by ≤ 0.022 N m (within scatter) and M_{ff} by ≤ 0.012 N m (0.39 mm of CoM offset).
4. *Family exclusivity.* The $d \rightarrow 0$ quadratic limit, fitted over the same windows, incurs 14–75% truncation error: the data select the cosh member, not merely any smooth monotone curve.

In summary: structurally, every modeled effect renames the effective constants (C_2, K) that the calibration estimates, so to first order the response cannot leave the cosh family; the sole escape channel is measured — not merely bounded — at a few mN m and propagates to a sub-1%, sub-noise shape deviation. Empirically, the zero-freedom family predicts the response across all runs with a noise-dominated, rate-invariant residual, is insensitive to the window extent, and decisively outperforms its own quadratic limit. The theoretical bounds serve as conservative a priori envelopes; every headline figure is a measurement.